

Command & Conquer(tm) Generals

STATUS REPORT FROM THE BATTLEFIELD

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February 11, 2003

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- - Incoming transmission - -

General,

Victory against our enemies is almost at hand, but constantly changing battlefield conditions require that we remain ever vigilant. Our forces have compiled a list of issues that you need to be aware of if we hope to achieve our ultimate goal of global supremacy.

Good luck.

- - End transmission - -

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1. WINDOWS 98/98SE/ME/2000/XP

1.1. Microsoft DirectX:

Command & Conquer(tm) Generals is a Microsoft DirectX application. Version 8.1 of Microsoft DirectX is included on the CD and you have the option to install it when Command & Conquer Generals is installed.

2. TROUBLE SHOOTING

2.1. Before You Start:

Before running Command & Conquer Generals, be sure that you have the latest drivers for your video and audio cards installed onto your PC. You can download the appropriate drivers from your video and audio card manufacturer's Web site.

Be sure that you have disabled anti-aliasing in your Windows display properties before running Command & Conquer Generals. Leaving this feature enabled might conflict with some of the game's effects.

Do not run any other programs in the background, as this may conflict with Command & Conquer Generals. This includes any virus-protection program that you have running.

If you should experience problems running Command & Conquer Generals after changing something in the game's Options, you can always revert back to the default settings by deleting the "options.ini" file. If you're running Windows XP, you can find this file by double-clicking "My Computer" icon, opening the "My Documents" folder, and then opening the "Command and Conquer Data" directory.

If you're running Windows 98, 98SE, ME, or 2000, simply double-click on the "My Documents" folder located on your Windows desktop, and then open the folder named "Command and Conquer Generals Data." The "options.ini" file will be located in there.

2.2. The Minimum System Spec:

The minimum required system specifications to run Command & Conquer Generals is as follows:

OS: Windows XP/ME/2000/98 (Windows 95/NT not supported)

CPU: 800 MHz Intel Pentium III or AMD Athlon processor

RAM: 128 MB

CD/DVD-Rom Speed: 8x

Hard Drive Space: 1.8 GB free hard disk space plus space for saved games, Windows swap-file, and DirectX 8.1

Video: 32 MB AGP video card using the Nvidia GeForce 2, ATI Radeon 7500, or more recent chipset with DirectX 8.1 compatible driver

Sound: DirectX 8.1 compatible PCI 16-bit sound card

Input: Keyboard, Mouse

Additionally, the following specifications are required for multiplayer:

256 MB RAM for 3-8 players

One set of game discs per player. Disc 1 and Disc 2 cannot be used simultaneously for multiplayer games.

Internet (2-4 players): 56Kbps or faster direct Internet connection

Internet (5-8 players): Cable, DSL, or a faster direct Internet connection

Network (2-8 players): TCP/IP compliant network

Recommended system specifications:

CPU: 1.8 GHz or faster Intel Pentium IV or AMD Athlon processor

RAM: 256 MB or more

Video: Nvidia GeForce3 or better Direct3D capable video card

Note: Some PCs equipped with the VIA KT133a chipset might encounter problems with Command & Conquer Generals.

2.3. Known Video Card Issues:

For optimum video performance, we highly recommend that you run Command & Conquer Generals with an AGP video card. PCI video cards are generally much slower than AGP cards, and you may experience abnormally slow performance and poor graphics with a PCI-based graphics card. Additionally, the video cards listed below are below our minimum supported specs.

Leadtek Winfast GeForce 2 GTS (AGP, 32MB)

SiS Xabre (AGP, 64MB)

Gainward Xabre 400 (AGP, 64MB)

Additionally, Some Nvidia GeForce 2 MX owners may experience some stuttering problems during movies.

If you attempt to run Command & Conquer Generals at 1600x1200 with a 32MB Nvidia GeForce 2 video card, the game may crash to the desktop. If this happens, try reducing the game's resolution.

Under all resolutions, you may experience that some objects like trees, walls, and pillars will flicker briefly. If this happens, saving and reloading your game will resolve the issue.

2.4. Known Sound Card Issues:

Before running Command & Conquer Generals, be sure that you have your sound card's latest drivers installed onto your PC. You can download the appropriate audio drivers from your sound card manufacturer's Web site.

Command & Conquer Generals does not support any ISA sound cards. If you own an ISA sound card, you will experience stuttering or dropped sound effects as well as frequent pauses during gameplay.

2.5. Virtual Memory Settings:

Command & Conquer Generals may need to use more RAM than is present on your system. Windows will automatically take care of this by using what is known as "Virtual Memory," which uses space on your hard disk to simulate the memory it needs and swaps data back and forth from your hard disk as required. Ensure that the maximum setting is at least 512MB.

2.6. AutoPlay:

If the Command & Conquer Generals CD does not AutoPlay when inserted in the drive, you may have this feature disabled. To enable AutoPlay, please consult your Windows user manual.

If you are running Windows XP, AutoPlay should be always enabled. You can customize your AutoPlay options by opening "My Computer," and right clicking on your CD-ROM or DVD Drive icon and choosing "Properties," and then selecting the "AutoPlay" tab. If AutoPlay doesn't work properly, please consult your Windows XP user manual.

2.7. East Asian Fonts:

All players who use a Western version of Windows will see boxes in place of Asian text in the Command & Conquer Generals online lobby. If you are running Windows XP, you can install the Asian language font pack by opening your Control Panel in Windows, double-clicking on the "Regional and Language Options" icon, selecting the "Languages" tab, and checking the "Install files for East Asian Languages" box. You will be asked for your Windows XP CD, and you will need to restart your PC. Once this process has been done, be sure that the "Asian Text" option has been selected under your "Persona" screen in Command & Conquer Generals' "Online" menu.

2.8. Game Updates and Patches:

Command & Conquer Generals contains an auto-update feature that will automatically let you know when new patches and updates are available to you. When an update to the game is available, you will be prompted to download and install it when you click on "Multiplayer" from the game's main menu. If you don't intend to play Generals online, but still wish to download the latest updates, you can either log onto the official Generals Web site at generals.ea.com or choose the "Check for Updates" option from the Command & Conquer Generals AutoRun feature.

2.9. Firewall Issues:

If you are running a firewall (particularly a NETGEAR solution) and are unable to connect to other players while in the Command & Conquer Generals multiplayer lobby, check the Send Delay box under the Network Options portion of the game's Options menu. If you are still experiencing connection problems, you may want to place your firewall in DMZ mode. (WARNING: Doing so will open your firewall ports to everyone, and this should only be done as a last resort). If you do not know how to configure your firewall settings, consult your firewall documentation or contact your Internet service provider.

If you have made changes to your firewall configuration since the last time that you played Command & Conquer Generals online and are experiencing connection difficulties, click the Refresh NAT box under the Network Options portions of the game's Option menu. It is highly recommended that you click this box every time you change your Internet or firewall settings.

2.10. Firewall/NAT Port Number Info:

The game will need to talk to external servers that use the following port numbers. These ports need to be open in your firewall in order to play on Generals Online:

TCP ports:

80, 6667, 28910, 29900, 29920

UDP ports:

4321, 27900

If your firewall does not allow outgoing packets to open ports to incoming traffic, or you experience problems connecting to other players, you will have to specify which port you want Command & Conquer Generals to use for communicating with other players. To do this, perform the following steps.

Go to your "My Documents" directory, and go to the "Command and Conquer Generals Data" directory. Right click on the options.ini file and choose "Edit."

Look for the "FirewallPortOverride" entry. If you don't see this entry, add it as a new line. Then assign the port you want to use to the port override entry. After making this change, this entry should look like:

```
FirewallPortOverride = XXXX
```

Where XXXX is the port number you have chosen. You will need to open the chosen port in your firewall for communication with players external to your firewall or NAT device. If you use the port override feature, the port you have chosen can not be masqueraded by a NAT device.

2.11. Uninstalling:

If you encounter any problems reinstalling Command & Conquer Generals after the game has been uninstalled, insert the first game disk into your CD-ROM drive, choose "Install" from the splash screen, and then select the "Repair" option. Once this process has been completed, install the game once again.

2.12. Windows XP Issues:

Limited user accounts are not supported by Command & Conquer Generals. You must have administrator access to install or uninstall the game. If you are playing under a Limited User account, you must use the Run as Administrator option.

3. ADVANCED DISPLAY OPTIONS

3.1. Advanced Options Explained

If you are experiencing slow framerate during gameplay, you can try adjusting one or several of the Advanced Display Options under the Options menu in Command & Conquer Generals. To access the Advanced Display Options, choose the Custom option from the Detail pulldown menu. Turning off the following options will improve game performance.

3D Shadows: This toggles showing the dynamic 3D shadows that units and buildings in the game.

2D Shadows: This toggles showing 2D shadows in the game.

Cloud Shadows: This toggles the appearance of moving cloud shadows on the ground in the game.

Extra Ground Lighting: This toggles the appearance of detailed lighting on the ground in the game.

Smooth Water Borders: This toggles smooth water borders along the shores in the game.

Behind Building: This toggles the silhouettes of any units that are standing or moving behind structures.

Show Props: This toggles the appearance of trees and shrubs that are scattered throughout the various levels in the game.

Extra Animations: This toggles extra effects like tree-swaying motions in the game.

Disable Dynamic LOD: This toggles the automatic adjustment of the game's detail level. Leave this unchecked for optimum performance.

Texture Resolution Slider: Moving the slider to the left will reduce the visual clarity of the textures in the game, but it will also improve performance.

Particle Cap Slider: Moving the slider to the left will reduce the amount of particle effects, like explosions and dust clouds, that appear in the game, but it will also improve performance.

4. OTHER INFO

4.1. EA Tech Support:

If you have questions about the program or trouble with Command & Conquer Generals, please refer to the user manual. Command & Conquer Generals also includes the EA Help file that provides solutions and answers to the most common difficulties and questions about how to properly use this product. It also allows you to send a detailed report of your problem directly to EA Technical Support via e-mail.

To access the EA Help file:

Start the Windows operating system.

Insert the Command & Conquer Generals disc into your CD-ROM/DVD-ROM drive. The AutoRun menu appears.

If the AutoRun menu does not automatically appear, left-click the "Start" button from the Windows Taskbar and select Run.... Type "D:\autorun.exe" in the Run dialog box, then click OK (substitute the correct letter of your CD-ROM/DVD-ROM drive if other than 'D:').

Click the Help button on the AutoRun menu. EA Help opens.

If you are still experiencing difficulty after utilizing the information in the EA Help file, you can also use it to contact EA Technical Support. EA Help contains a utility that detects your system's hardware and organizes this information into a detailed report. This report helps EA Technical Support solve your problem in the quickest possible time. Please note that you will need to have registered your copy of Command & Conquer Generals in order to take advantage of this utility.

To run the utility:

Launch the EA Help file.

Go to page 2 of the Welcome screen by clicking the small arrow in the bottom corner.

Click the Contact Technical Support button. The utility opens.

Click OK to begin detecting your hardware.

When it finishes, you can view the gathered information by looking through the various category tabs. To send this information to EA Technical Support, fill out the blank sections under the User tab including a detailed description of the problem you are experiencing and click Send (if you have an open Internet connection) or Save Info (if you wish to e-mail the file manually). A copy of the report is automatically saved to your Windows desktop.

4.2. Gameplay Changes:

The user manual incorrectly states that players will earn one point for every match lost while playing Command & Conquer Generals online. If you lose a match while playing against opponents online, you will not earn any points. If you win a match, you will earn three points.